



B.4 Thermodynamics (HL)

The particulate nature of matter · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The first law of thermodynamics states that the change in internal energy of a system equals the heat supplied to it plus the work done on it, $\Delta U = Q + W$ (sign convention: W positive when done on the gas). For an ideal gas, internal energy depends only on temperature, so an isothermal process has $\Delta U = 0$; an adiabatic process has $Q = 0$ (no heat exchange); an isovolumetric (isochoric) process has zero work done since $\Delta V = 0$; and an isobaric process occurs at constant pressure with $W = p\Delta V$. The second law of thermodynamics implies that in any real (irreversible) process the entropy of an isolated system increases, meaning no engine can convert heat entirely into work with 100% efficiency; a Carnot engine sets the theoretical maximum efficiency, $\eta_{\text{max}} = 1 - T_{\text{cold}}/T_{\text{hot}}$, using absolute temperatures.

Worked example: follow the reasoning

An ideal gas absorbs 500 J of heat and does 200 J of work on its surroundings while expanding. Find the change in internal energy.

Sign convention: $\Delta U = Q + W$, where W is work done ON the gas. The gas does 200 J of work on surroundings, so work done on the gas is $W = -200$ J.

$\Delta U = Q + W = 500 + (-200) = 300$ J.

Answer: $\Delta U = +300$ J (internal energy increases by 300 J).

Quick check · multiple choice

A heat engine operates between a hot reservoir at 500 K and a cold reservoir at 300 K. What is its maximum possible (Carnot) efficiency?

- A 20%
- B 40%
- C 60%
- D 80%

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A fixed mass of ideal gas undergoes an isothermal expansion at 320 K, absorbing 850 J of heat from its surroundings. [7]
 - (a) State the value of ΔU for this process and explain why.
 - (b) Calculate the work done by the gas during this expansion.
 - (c) A heat engine operating between reservoirs at 320 K and 260 K is later proposed to use this gas. Calculate its maximum theoretical (Carnot) efficiency and explain why real engines always have lower efficiency than this value.

Final answer: _____



Answers and explanation

Question 1 [7]

- (a) $\Delta U = 0$, because for an ideal gas internal energy depends only on temperature, and temperature is constant in an isothermal process.
- (b) From the first law, $\Delta U = Q + W(\text{on gas}) \rightarrow 0 = 850 + W(\text{on gas}) \rightarrow W(\text{on gas}) = -850 \text{ J}$, so work done BY the gas = +850 J.
- (c) $\eta_{\text{max}} = 1 - T_c/T_h = 1 - 260/320 = 1 - 0.8125 = 0.1875 \approx 18.8\%$. Real engines have irreversible losses (friction, non-quasi-static heat transfer, turbulence) that generate entropy, so their efficiency is always below this theoretical reversible-engine maximum.

Quick check answer

B. $\eta_{\text{max}} = 1 - T_c/T_h = 1 - 300/500 = 1 - 0.60 = 0.40 = 40\%$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.